THURSDAY, April 30

ex 23

Claim: If $\lim_{i\to\infty} a_i = \lim_{i\to\infty} b_i = L$ then $(a_i) \sim (b_i)$.

WTS: $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$: if $i > N$, then $|a_i - L| < \varepsilon$.

scratch →

$|a_i - L|$, $|b_i - L|$ can be small b/c they converge → want $a_i - b_i$ to be small

trick 1 add & subtract something

$|a_i - b_i| = |a_i - L + L - b_i|$

triangle inequality

$\leq |a_i - L| + |L - b_i| < \varepsilon$

$\frac{\varepsilon}{2}$

Pf

Let $\varepsilon > 0$. Since $\lim_{i\to\infty} a_i = \lim_{i\to\infty} b_i = L$ we can find N_1 & $N_2 \in \mathbb{N}$: if $i \geq N_1$ then $|a_i - L| < \varepsilon/2$ & if $i \geq N_2$, the $|b_i - L| < \varepsilon/2$.

Let $N = \max(N_1, N_2)$. If $i \geq N$, then

$|a_i - b_i| = |a_i - L + L - b_i| \leq |a_i - L| + |L - b_i| \leq |a_i - L| + |b_i - L|$
$< \varepsilon/2 + \varepsilon/2 < \varepsilon$.

Where the first inequality comes from the triangle inequality.

$\therefore a_i \sim b_i$. ∎

Problem 1:

$\sqrt{2}$ & friends $\notin \mathbb{Q}$.

Solution: take sequences of rationals.

$1/10,\ 141/100,\ 1414/1000,\ \ldots\ \sqrt{2}$

Problem 2:

There are multiple sequences that approach the same thing.

$1,\ \frac{14}{10},\ \frac{14}{100},\ \frac{1414}{1000},\ \ldots,\ \sqrt{2}$

$1,\ 1+\frac{1}{3},\ 1+\frac{1}{3}+\frac{2}{27},\ 1+\frac{1}{3}+\frac{2}{27}+\frac{2}{33},\ \ldots\ \sqrt{2}$

Solution: take equivalence classes of sequences

$(a_i) \sim (b_i)$ $\overset{\text{def}}{\text{if}}$ $\lim_{i\to\infty} a_i - b_i = 0$

Problem 3: There are sequences that shouldn't converge.

$1, 2, 3, 4, 5, \ldots$ $\qquad$ $1, -1, 1, -1, 1, \ldots$

Solution: only consider equivalence classes of sequences that "should" converge.

Define a sequence a_i to be <u>Cauchy</u>. If $\forall\ \varepsilon > 0\ \exists\ N \in \mathbb{N}$ such that if $i, j \geq N$, $|a_i - a_j| < \varepsilon$.

CHAPTER 10

<u>def (1) & (2)</u> → the Real Numbers ($\mathbb{R}$)

If a_i is a cauchy sequence in $\mathbb{Q}$, then define $[(a_i)]$ to be the equivalence class of (a_i). I.e $[(a_i)] = \{(b_i) : (b_i) \sim (a_i)\}$

Define the real numbers ($\mathbb{R}$) to be $\mathbb{R} \overset{\text{def}}{=} \{[a_i]] : (a_i)$ is a Cauchy sequence in $\mathbb{Q}$.

<u>def 3</u> ($+, \times$ on $\mathbb{R}$)

$$[(a_i)_{i \geq n}] + [(b_i)_{i \geq m_0}] = [(a_i + b_i)_{i \geq \max(n_0, m_0)}] \text{ add.}$$

$$[(a_i)_{i \geq n}] - [(b_i)_{i \geq m_0}] = [(a_i + b_i)_{i \geq \max(n_0, m_0)}] \text{ mult.}$$

Do operations w/sequences work out well?

— Chp. 8 —

Thm. 44 — Suppose $a_i = a\ \forall i$. Then $\lim_{i \to \infty} a_i = a$.

Thm. 45 — If $a_i \leq b_i$, $\forall i$ (sufficiently large), then $\lim_{i \to \infty} a_i \leq \lim_{i \to \infty} b_i$.

⋮

Calc 1 limit laws foreverrrr....

CAUCHY Pf:

Define a sequence (a_i) to be Cauchy if $\forall\ \varepsilon > 0\ \exists\ N \in \mathbb{N}$ such that if $i, j > N$, $|a_i - a_j| < \varepsilon$.

Chp. 9, thm. 15 → Suppose $\lim_{i\to\infty} a_i = L$. Then (a_i) is Cauchy.

Pf
Let $\varepsilon > 0$. Since $\lim_{i\to\infty} a_i = L$ $\exists\ N \in \mathbb{N}$: if $i > N$, then $|a_i - L| < \frac{\varepsilon}{2}$.
If $i, j \geq N$, then

$$|a_i - a_j| = |a_i - L + L - a_j| \overset{\Delta \text{ inequality}}{\leq} |a_i - L| + |L - a_j| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \leq \varepsilon.$$

$+L - L$ trick

Thus a_i is Cauchy. $\boxed{D}$

$\boxed{\text{Thm. 16}}$ Chp. 9

If $(a_i) \sim (b_i)$ & (a_i) is Cauchy, then (b_i) is Cauchy.

- have control of a_i & b_i getting closer
- " " of a_i's terms getting closer

Sketch
WTS → $|b_i - b_j|$ to be small
know: $|a_i - a_j|$ is small
: $|a_i - b_i|$ is small

$$|b_i - b_j| = |b_i - a_i + a_i - a_j + a_j - b_j| \leq \underbrace{|b_i - a_i|}_{\varepsilon/3} + \underbrace{|a_i - a_j|}_{\varepsilon/3} + \underbrace{|a_j - b_j|}_{\varepsilon/3}$$

Pf
Let $\varepsilon > 0$. Since $(a_i) \sim (b_i)$ & (a_i) is Cauchy, $\exists$ $N_1 \in \mathbb{N}$: if $i > N$ then $|a_i - b_i| < \varepsilon/3$. And $\exists$ $N_2 \in \mathbb{N}$: if $i, j \geq N_2$, $|a_i - a_j| < \varepsilon/3$.
Let $N = \max(N_1, N_2)$. If $i, j \geq N$, then:

$$|b_i - b_j| = |b_i - a_i + a_i - a_j + a_j - b_i| \leq |b_i - a_i| + |a_i - a_j| + |a_j - b_j| < \varepsilon/3 + \varepsilon/3 + \varepsilon/3 < \varepsilon.$$

Thus (b_j) is Cauchy! $\square$